

Power & TIE plots

2026-03-12

```
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

options(scipen = 999)

# Load data ----
setwd("/Users/remus/Documents/GitHub/latent_interaction/sim_latent_interaction")
load(file = "agg_results.rda")

# Get unique combinations of rsq_prod and n_items
combinations <- agg_results %>%
  distinct(rsq_prod, n_items)

# Loop over each combination

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for (i in 1:nrow(combinations)) {

  plot_data <- agg_results %>%
    filter(rsq_prod == combinations$rsq_prod[i], n_items == combinations$n_items[i],
           group_prob %in% c(1,2))

  # Skip if no data
  if (nrow(plot_data) == 0) next

  plot_data <- plot_data %>%
    mutate(
      group_prob = factor(group_prob, levels = c(1, 2)),
      loading     = factor(loading),
      model_type  = factor(model_type),
```

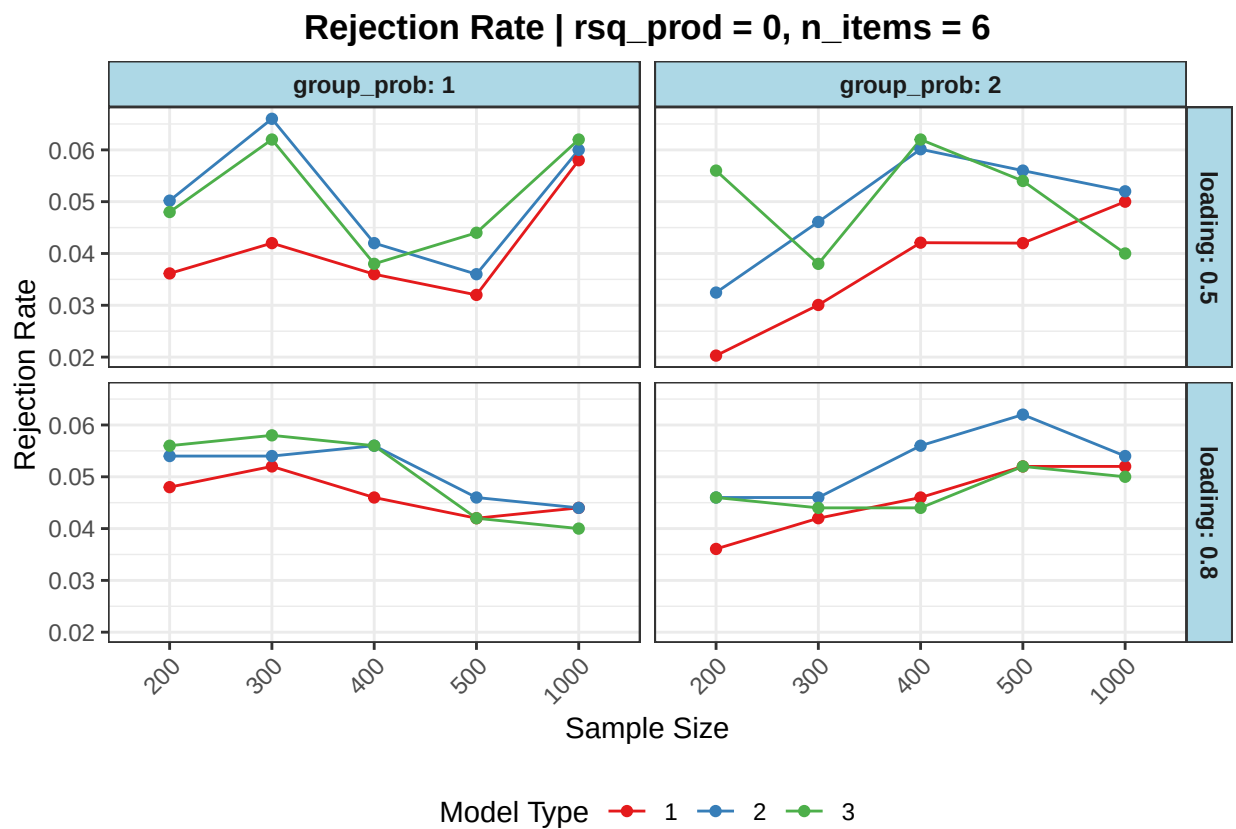
```

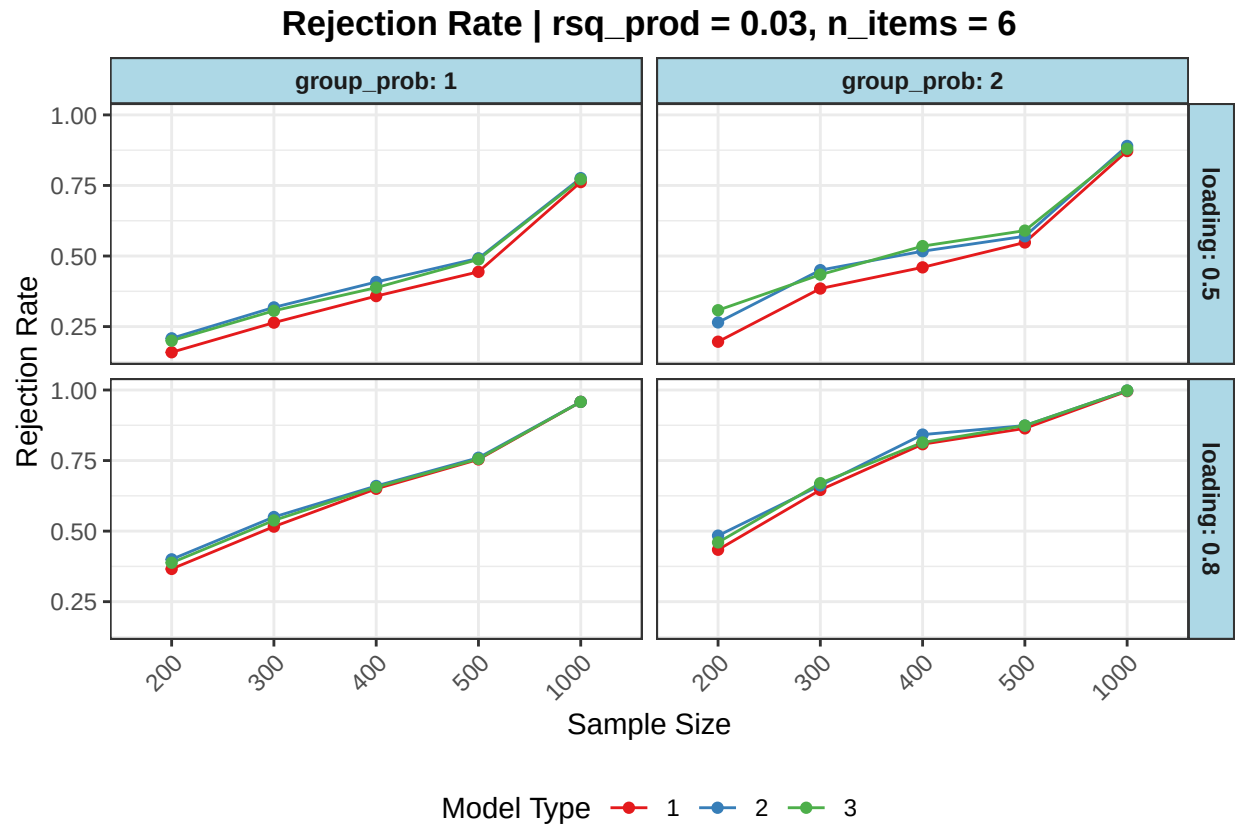
sample_size = factor(sample_size))

p <- ggplot(plot_data, aes(x = sample_size, y = sig,
                           color = model_type,
                           group = model_type)) +
  geom_line() + geom_point() + facet_grid(rows = vars(loading),
                                           cols = vars(group_prob),
                                           labeller = label_both) +
  scale_color_brewer(palette = "Set1", name = "Model Type") +
  labs(
    title = paste0("Rejection Rate | rsq_prod = ", combinations$rsq_prod[i],
                  ", n_items = ", combinations$n_items[i]),
    x = "Sample Size", y = "Rejection Rate") +
  theme_bw() +
  theme(
    plot.title       = element_text(hjust = 0.5, face = "bold"),
    legend.position  = "bottom",
    axis.text.x      = element_text(angle = 45, hjust = 1),
    strip.background = element_rect(fill = "lightblue"),
    strip.text       = element_text(face = "bold"))

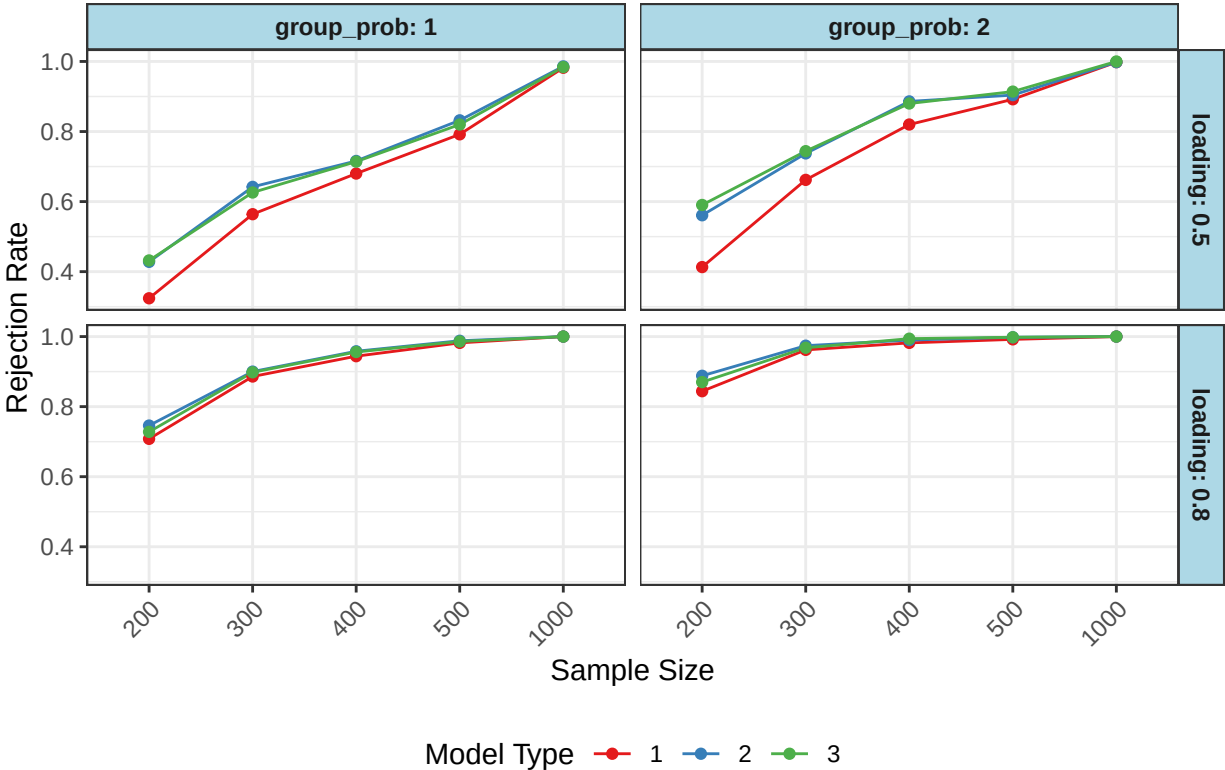
print(p)
}

```

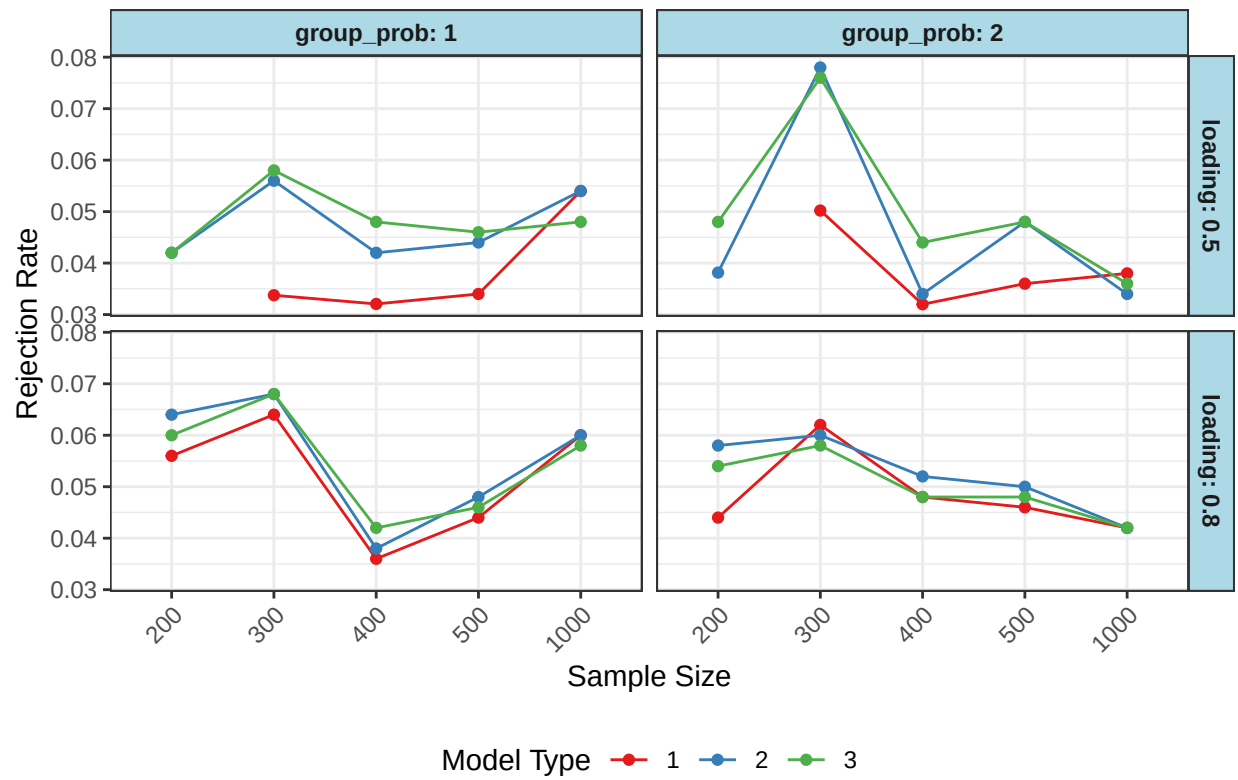




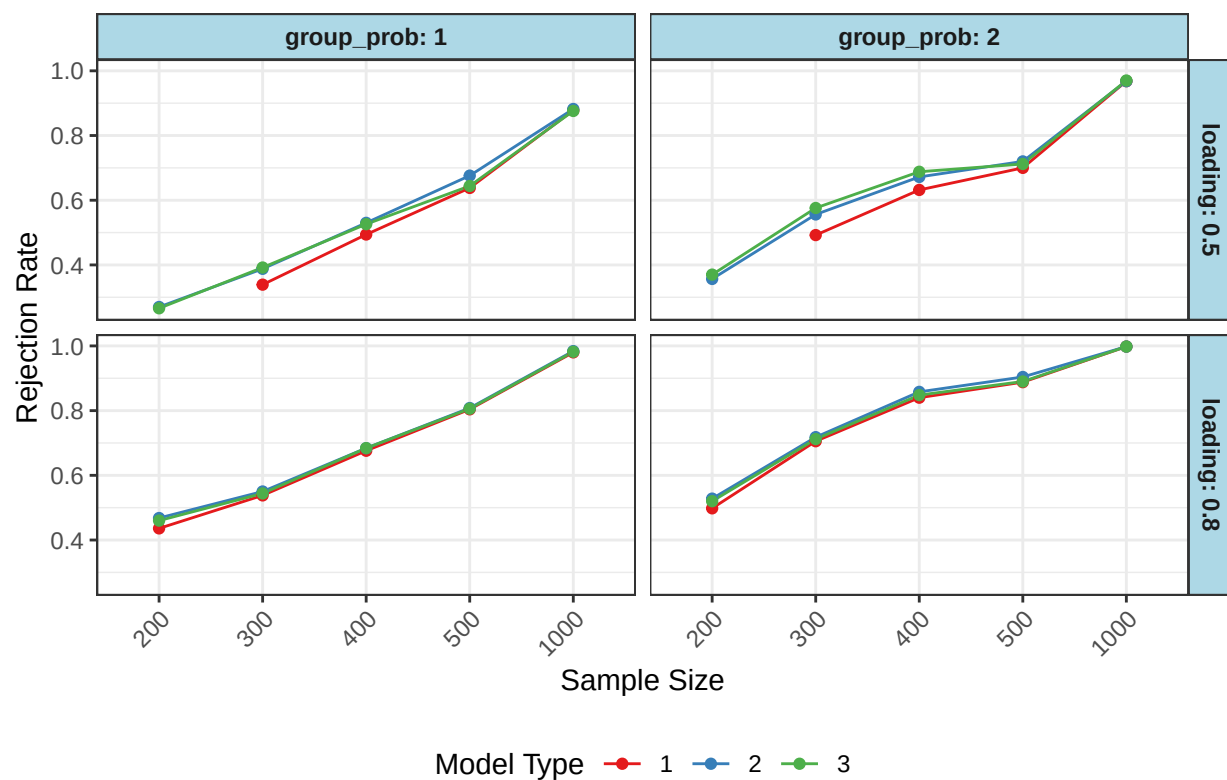
Rejection Rate | $rsq_prod = 0.07$, $n_items = 6$



Rejection Rate | $rsq_prod = 0$, $n_items = 12$



Rejection Rate | $rsq_prod = 0.03$, $n_items = 12$



Rejection Rate | $rsq_prod = 0.07$, $n_items = 12$

